

Assignment # 1

Semester:Fall 2022

General Science (GSC101)

Total Marks: 20

Due Date: 6th June,2022

Objective:

The assignment has been designed to enhance your knowledge and understanding about Laws of Physics and application of Physics in our daily life.

Instructions:

- Please read the following instructions carefully before solving & submitting assignment:
- To solve this assignment, you should have good command over first **10 lectures**.
- The assignment will not be accepted after due date.
- Zero marks will be awarded to the assignment that does not open or the file is corrupt.
- The assignment file must be an MS word (doc) file format; Assignment will not be accepted in any other format.
- Use Math Type or Equation Editor etc for mathematical symbols.
- **Zero marks will be awarded to the assignment if copied (from other student or copied from handouts or internet).**
- Don't wait for the last date to submit your assignment.

For any query about the assignment, contact only gsc101@vu.edu.pk

Do not post queries related to assignment on MDB.

[illegible]

Question-1**(5 Marks)**

Define the term of efficiency and write its formula. Does it have a unit or dimension? Briefly describe your answer with examples? (3+1+1)

Definition

Useful energy output to the input is called efficiency. or
Fraction of energy of a system, performing useful work, is the efficiency of that system.

Formula:

$$\text{Efficiency}\% = \frac{E_{\text{output}}}{E_{\text{input}}} \times 100\%$$

Unit and Dimension:

Efficiency is a unit less quantity. It is the ratio of the energy output to the energy input; hence it has no units and Dimension.

Question-2**(7 Marks)**

A hair dryer releases 1 KJ of energy in half a minute from the effect of 0.1 kW power given to the system. What is the maximum efficiency of a hair dryer? Is it possible that the efficiency of hair dryer is 100%? Explain your answer. (5+1+1)

Answer:

$$\text{Time} = t = \frac{1}{2} \text{ hour} = 30 \text{ sec}$$

$$P_{\text{input}} = 0.1 \text{ kW}$$

$$P_{\text{output}} = \frac{E}{t} = \frac{1 \text{ kJ}}{30 \text{ sec}} = \frac{1}{30} \text{ kW}$$

$$\text{Efficiency} = \frac{E_{\text{output}}}{E_{\text{input}}} \times 100\%$$

$$\text{Efficiency} = \frac{\frac{1}{30} \text{ kW}}{0.1 \text{ kJ}} \times 100\%$$

$$\text{Efficiency} = \frac{1 \text{ kW}}{30 \times 0.1 \text{ kJ}} \times 100\%$$

$$\text{Efficiency} = \frac{1 \text{ kW}}{30 \text{ kW}} \times 100\%$$

$$\text{Efficiency} = 33.3\%$$

Yes, it is possible that the efficiency of hair dryer is 100%. It depends upon what portion of the electricity use is for the heating element, and what portion is used for the fan. The Heating element is a resistive load, and is 100% efficient. The fan is an inductive load, and is not 100% efficient. I imagine that in most hairdryers they are not putting super high-efficiency motors in to drive the fans, but I also suspect that the vast majority of the energy used will be by the heating coils, so your overall efficiency will likely still be very high.

heat engine

In a heat engine, heat is transferred from a higher temperature level called source to a lower temperature level called sink. Work is obtained during this process.

Carnot engine

It is an ideal heat engine whose efficiency is maximum. In Carnot engine, Carnot cycle is used. In Carnot engine, ideal gas is the working substance and all the dissipative forces are absent.

Steam engines and turbines operate on the Rankin cycle which has a maximum Carnot efficiency of 63% for practical engines, with steam turbine power plants able to achieve efficiency in the mid 40% range.

The efficiency of steam engines is primarily related to the steam temperature and pressure and the number of stages or expansions. Steam engine efficiency improved as the operating principles were discovered, which led to the development of the science of thermodynamics

In a steam engine, hot steam, usually supplied by a boiler, expands under pressure, and part of the heat energy is converted into work. The remainder of the heat may be allowed to escape, or, for maximum engine efficiency